

REAL-TIME BALL TRACKING (Using Only the Phone Camera)

This is possible with simple computer vision.

Your app can detect:

- Ball speed
- Line & length
- Swing direction
- Pitch point
- Whether shot selection was correct

This creates a full **SmartBat + SmartCam** ecosystem — no one has this.

AI “SHOT SUGGESTION ENGINE” — Like a Cricket Coach Brain

You already have:

- Fielding presets
- AI pose tracking
- Swing + timing data

Now combine these to make a **Shot Recommendation AI**:

Example:

“For this length + your position + your swing angle → recommended shot: Cover drive.”

This is **CRAZY ADVANCED** — beyond every competitor globally.

AFTER-SHOT “SHOULD YOU HAVE PLAYED THIS SHOT?” ANALYSIS

This is next level.

System checks:

- Ball length
- Field setup
- Timing

- Sweet spot accuracy
- Weight transfer
- Swing angle
- Bat speed

And says:

- ✓ Right shot
- ✗ Wrong shot (“Risky shot for this field”)
- ✓ Better option: Square cut

This is like giving every player their own **AI Rahul Dravid**.

AIR-GESTURE VOICE FEEDBACK

Instead of looking at phone:

SmartBat speaks:

- “Too early.”
- “Too late.”
- “Perfect middle!”
- “Swing angle low.”
- “Feet not aligned.”

This is **a real digital cricket coach**.

AI COACH – Session Review & 10-Min Personalized Guidance Video”

(This can become the signature feature of SmartBat)

What it does:

After a player finishes batting:

- The SmartBat app automatically processes the data
- Identifies patterns
- Understands mistakes
- Finds your strongest shot or area
- And finally... **generates a personalised 10-minute AI coaching video**

This video will explain:

- ✓ **Your timing mistakes**
- ✓ **Your sweet spot usage**
- ✓ **Your swing efficiency**
- ✓ **Your footwork based on camera AI**
- ✓ **What shot you should prefer**
- ✓ **How to correct your stance**
- ✓ **How to improve in the next session**

This becomes like having **Rahul Dravid, Ricky Ponting, or Smriti Mandhana** analyzing your session.

How It Works Internally (Simple Explanation for Judges)

Your app will:

1. Collect session data (IMU + piezo + camera)
2. Send data to your AI model
3. AI analyses mistakes + patterns
4. AI generates a scripted coaching explanation
5. Video is created using:

- Animated bat diagrams
- Overlays
- Pose skeleton highlights
- Voiceover (text to speech)
- Simple shot clips

This is all 100% possible with:

- MediaPipe
- Flutter overlays
- Firebase
- Simple TTS engines
- Pre-made coaching templates

PRESSURE SIMULATION MODE (Your idea → upgraded)

 ***“MATCH PRESSURE ENGINE”***

You already said:

15 runs in 6 balls, set field, feel pressure

Let's make it elite:

How it works

- **User selects scenario:**
 - **12 needed off 4**
 - **30 off 12**
 - **Save the wicket**
- **App auto-sets:**

- **Fielding presets**
- **Bowler type (pace/spin)**
- **Line & length probability**
- **AI evaluates decision quality, not just runs**

Feedback

- **“You survived pressure well”**
- **“Shot selection risky under pressure”**
- **“Better option: rotate strike”**

 **This becomes mental training, not just batting.**

 **Premium subscription feature**

FATIGUE & CONSISTENCY TRACKING (UNDERRATED)

Using IMU + session time:

- **Bat speed drop over session**
- **Timing degradation**
- **Shot quality under fatigue**

App says:

“Your timing drops after 42 balls.”

Coaches LOVE this.

